CURRICULUM VITAE

PERSONAL INFORMATION

Name:	**Yaqiang Cao**
Organization:	Systems Biology Center, National Heart, Lung, and Blood Institute (NHLBI), National Institutes of Health (NIH)
Work Address:	10 Center Drive, Room 6N248A, Bethesda, MD, 20892
Current Position:	Research Fellow
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h-index | i10-index:	22 | 26

EDUCATION

2019 **Ph.D. in Computational Biology**
Chinese Academy of Sciences (CAS)–Max Planck Society (MPG) Partner Institute for Computational Biology, University of Chinese Academy of Sciences (UCAS), Shanghai, China.
Mentor: Prof. Jing-Dong Jackie Han

2012 **B.S. in Bioengineering**
East China University of Science and Technology, Shanghai, China.

RESEARCH EXPERIENCE

2024-Curr. **Research Fellow**
Systems Biology Center, National Heart, Lung, and Blood Institute (NHLBI), National Institutes of Health (NIH), Bethesda, MD, USA.
Mentor: Dr. Keji Zhao

2019-2024 **Visiting Fellow**
NHLBI/NIH, Bethesda, MD, USA.
Mentor: Dr. Keji Zhao

2011-2018 **Graduate Research Assistant**
CAS-MPG Partner Institute for Computational Biology, Shanghai, China.
Mentor: Prof. Jing-Dong Jackie Han

RESEARCH SUMMARY

Focus:	Chromatin regulation in immunity, leukemia, and aging.
Methods:	Developed computational and experimental methods for high-resolution enhancer–promoter and single-cell nucleosome mapping.
Discoveries:	Roles of chromatin loops, histone modifications, and transcription factors in T-cell differentiation, acute myeloid leukemia, and B-cell aging.
Partnerships:	Cross-disciplinary collaborations among biologists, physicians, and data scientists connecting basic research with clinical applications.
Impact:	User-friendly software for epigenomic and 3D genome research; mechanistic insights into chromatin regulation and its clinical implications.

FIRST/CO-FIRST PUBLICATIONS (16 out of 36)

[*] Equal contribution; [#] Corresponding author.

2026 ***Cao, Y.***[*], Ge, G.[*], & Zhao, K.[#] (2026). Ryder: Epigenome normalization using a two-tier model and internal reference regions. ***Bioinformatics***, btag753. Advance online publication. https://doi.org/10.1093/bioinformatics/btag753

Cao, Y., Tang, Q., & Zhao, K.[#] (2026). Elucidation of chromatin loops among regulatory regions using Hi-TrAC. ***Frontiers in Epigenetics and Epigenomics***, 4, 1858923. https://doi.org/10.3389/freae.2026.1858923

2025 Zhen, T.[*], ***Cao, Y.***[*], Dou, T., Chen, Y., Lopez, G., Menezes, A. C., Wu, X., Hammer, J. A., Cheng, J., Garrett, L., Anderson, S., Kirby, M., Wincovitch, S., Sisay, B., Elkahloun, A. G., Wu, D., Castilla, L. H., Yang, W., Jiang, J., Zhao, K., & Liu, P. P.[#] (2025). CBFβ-SMMHC–driven leukemogenesis requires enhanced RUNX1-DNA binding affinity in mice. ***The Journal of Clinical Investigation***, 135(19), e192923. https://doi.org/10.1172/JCI192923

2024 Ma, F.[*], ***Cao, Y.***[*], Du, H., Braikia, F. Z., Zong, L., Ollikainen, N., Bayer, M., Qiu, X., Park, B., Roy, R., Nandi, S., Sarantopoulou, D., Ziman, A., Bianchi, A. H., Beerman, I., Zhao, K., Grosschedl, R., & Sen, R.[#] (2024). Three-dimensional chromatin reorganization regulates B cell development during ageing. ***Nature Cell Biology***, 26(6), 991–1002. https://doi.org/10.1038/s41556-024-01424-9

Liu, S.[*], ***Cao, Y.***[*], Cui, K.[*], Ren, G., Zhao, T., Wang, X., Wei, D., Chen, Z., Gurram, R. K., Liu, C., Wu, C., Zhu, J., & Zhao, K.[#] (2024). Regulation of T helper cell differentiation by the interplay between histone modification and chromatin interaction. ***Immunity***, 57(5), 987–1004.e5. https://doi.org/10.1016/j.immuni.2024.03.018

2023 ***Cao, Y.***[*], Liu, S.[*], Cui, K.[*], Tang, Q., & Zhao, K.[#] (2023). Hi-TrAC detects active sub-TADs and reveals internal organizations of super-enhancers. ***Nucleic Acids Research***, 51(12), 6172–6189. https://doi.org/10.1093/nar/gkad378

Cui, K.[*], Chen, Z.[*], ***Cao, Y.***[*], Liu, S.[*], Ren, G., Hu, G., Fang, D., Wei, D., Liu, C., Zhu, J., Wu, C.[#], & Zhao, K.[#] (2023). Restraint of IFN-γ expression through a distal silencer CNS–28 for tissue homeostasis. ***Immunity***, 56(5), 944–958.e6. (**Cover article; highlighted in an Immunity Preview.**) https://doi.org/10.1016/j.immuni.2023.03.006

2022 ***Cao, Y.***[*], Liu, S.[*], Ren, G.[*], Tang, Q., & Zhao, K.[#] (2022). cLoops2: a full-stack comprehensive analytical tool for chromatin interactions. ***Nucleic Acids Research***, 50(1), 57–71. https://doi.org/10.1093/nar/gkab1233

Liu, S.[*], ***Cao, Y.***[*], Cui, K., Tang, Q., & Zhao, K.[#] (2022). Hi-TrAC reveals division of labor of transcription factors in organizing chromatin loops. ***Nature Communications***, 13(1), 6679. https://doi.org/10.1038/s41467-022-34276-8

2020 ***Cao, Y.***[*], Chen, Z.[*], Chen, X.[*], Ai, D.[*], Chen, G., McDermott, J., Huang, Y., Guo, X., & Han, J.-D. J.[#] (2020). Accurate loop calling for 3D genomic data with cLoops. ***Bioinformatics***, 36(3), 666–675. https://doi.org/10.1093/bioinformatics/btz651

Zhen, T.[*], ***Cao, Y.***[*], Ren, G., Zhao, L., Hyde, R. K., Lopez, G., Feng, D., Alemu, L., Zhao, K., & Liu, P. P.[#] (2020). RUNX1 and CBFβ-SMMHC transactivate target genes together in abnormal myeloid progenitors for leukemia development. ***Blood***, 136(21), 2373–2385. (**Plenary Paper**) https://doi.org/10.1182/blood.2020007747

Han, M.[*], Li, J.[*], ***Cao, Y.***[*], Huang, Y., Li, W., Zhu, H., Zhao, Q., Han, J.-D. J., Wu, Q., Li, J., Feng, J.[#], & Wong, J.[#] (2020). A role for LSH in facilitating DNA methylation by DNMT1 through enhancing UHRF1 chromatin association. ***Nucleic Acids Research***, 48(21), 12116–12134. https://doi.org/10.1093/nar/gkaa1003

Zeng, Y.[*], ***Cao, Y.***[*], Halevy, R. S.[*], Nguyen, P., Liu, D., Zhang, X., Ahituv, N.[#], & Han, J.-D. J.[#] (2020). Characterization of functional transposable element enhancers in acute myeloid leukemia. ***Science China Life Sciences***, 63(5), 675–687. https://doi.org/10.1007/s11427-019-1574-x

Huang, F.[*#], ***Cao, Y.***[*], Wu, G., Chen, J., Wang, C., Lin, W., Lan, R., Wu, B., Xie, X., Hong, J., & Fu, L. (2020). BMP2 signalling activation enhances bone metastases of non-small cell lung cancer. ***Journal of Cellular and Molecular Medicine***, 24(18), 10768–10784. https://doi.org/10.1111/jcmm.15702

2019 ***Cao, Y.***[*], Chen, G.[*], Wu, G.[*], Zhang, X.[*], McDermott, J., Chen, X., Xu, C., Jiang, Q., Chen, Z., Zeng, Y., Ai, D., Huang, Y., & Han, J.-D. J.[#] (2019). Widespread roles of enhancer-like transposable elements in cell identity and long-range genomic interactions. ***Genome Research***, 29(1), 40–52. https://doi.org/10.1101/gr.235747.118

2017 Wang, L.[*], Xu, X.[*], ***Cao, Y.***[*], Li, Z., Cheng, H., Zhu, G., Duan, F., Na, J., Han, J.-D. J., & Chen, Y.-G.[#] (2017). Activin/Smad2-induced histone H3 Lys-27 trimethylation (H3K27me3) reduction is crucial to initiate mesendoderm differentiation of human embryonic stem cells. ***Journal of Biological Chemistry***, 292(4), 1339–1350. https://doi.org/10.1074/jbc.M116.766949

OTHER PUBLICATIONS

2026 Zhu, X., Song, J., Nie, J., Chen, X., ***Cao, Y.***, Wei, D., Gurram, R. K., Peng, D., Zhao, K., Bosselut, R., & Zhu, J. (2026). The transcription factor Bhlhe40 promotes inflammatory functions of ILC2s through inducing GM-CSF while inhibiting IL-10 expression. ***bioRxiv*** [Preprint]. https://doi.org/10.64898/2026.08.23.746557

Liu, N., Xu, J., Ku, W. L., Chen, W., ***Cao, Y.***, Kazmi, R., Jin, W., Gauthier, T., Luo, S., Shen, S., Patimo Molano, L., Lim, Y.-J., Ottaviani, V., Naylor, E., Zhao, K., & Chen, W. (2026). Destabilization of intratumor Tregs by CTLA-4 engagement confers anti-CTLA-4-driven immunotherapy. ***bioRxiv*** [Preprint]. https://doi.org/10.64898/2026.07.28.741328

Khateb, M., Jung, R., Hadley, P., Leibou, S., Yu, Z., Dinerman, A. J., Dulemba, V., Gasmi, B., Levin, N., Kim, S. P., Bhasin, A., Bhat, D., Sindiri, S., Gartner, J. J., Prickett, T. D., Benzine, T., Farid, S., Parkhurst, M., Halas, H., Ge, G., ***Cao, Y.***, Zhao, K., Yang, J. C., Robbins, P., Lowery, F. J., Krishna, S., Heller, T., McVicar, D., Rosenberg, S. A., & Klemen, N. D. (2026). Selective expansion of neoantigen-specific peripheral blood lymphocytes through antigen-presenting cell-T cell conjugate isolation. ***Journal for ImmunoTherapy of Cancer***, 14(10), e014996. https://doi.org/10.1136/jitc-2026-014996

Zong, L., Park, B., ***Cao, Y.***, Ma, F., Tekin-Turhan, F., Kuribayashi, W., Zhao, K., & Beerman, I. (2026). Chromatin reorganization drives overexpression of a Btaf1 variant underpinning hematopoietic aging. ***Nature Communications***, 17(1), 4129. https://doi.org/10.1038/s41467-026-70787-4

2025 Zhu, X., Chen, X., ***Cao, Y.***, Liu, C., Kline, Z. J., Hu, G., Ganesan, S., Veres, T. Z., Fang, D., Liu, S., Wei, D., Shibata, H., Golec, D. P., Chung, H., Germain, R. N., Schwartzberg, P. L., Zhao, K., & Zhu, J. (2025). Optimal CXCR5 expression during Tfh maturation involves the Bhlhe40-Pou2af1 axis. ***Cell Reports***, 44(11), 116470. https://doi.org/10.1016/j.celrep.2025.116470

Roy, S., Ren, M., Li, P., Cui, K., ***Cao, Y.***, Fisk, B., Markowitz, T. E., Redekar, N., Sakamoto, K., Nagao, K., Oh, J., Spolski, R., Liao, W., Dubois, S. P., Kelsall, B. L., Zhao, K., Phelan, J. D., & Leonard, W. J. (2025). BLIMP1 negatively regulates IL-2 signaling in T cells. ***Science Advances***, 11(29), eadx8105. https://doi.org/10.1126/sciadv.adx8105

2022 Zhang, X., Jiang, Q., Li, J., Zhang, S., ***Cao, Y.***, Xia, X., Cai, D., Tan, J., Chen, J., & Han, J.-D. J. (2022). KCNQ1OT1 promotes genome-wide transposon repression by guiding RNA–DNA triplexes and HP1 binding. ***Nature Cell Biology***, 24(11), 1617–1629. https://doi.org/10.1038/s41556-022-01008-5

Lin, X., Liu, Y., Liu, S., Zhu, X., Wu, L., Zhu, Y., Zhao, D., Xu, X., Chemparathy, A., Wang, H., ***Cao, Y.***, Nakamura, M., Noordermeer, J. N., La Russa, M., Wong, W. H., Zhao, K., & Qi, L. S. (2022). Nested epistasis enhancer networks for robust genome regulation. ***Science***, 377(6610), 1077–1085. https://doi.org/10.1126/science.abk3512

Ren, G., Lai, B., Harly, C., Baek, S., Ding, Y., Zheng, M., ***Cao, Y.***, Cui, K., Yang, Y., Zhu, J., Hager, G. L., Bhandoola, A., & Zhao, K. (2022). Transcription factors TCF-1 and GATA3 are key factors for the epigenetic priming of early innate lymphoid progenitors toward distinct cell fates. ***Immunity***, 55(8), 1402–1413.e4. https://doi.org/10.1016/j.immuni.2022.06.019

Pan, L., Ku, W. L., Tang, Q., ***Cao, Y.***, & Zhao, K. (2022). scPCOR-seq enables co-profiling of chromatin occupancy and RNAs in single cells. ***Communications Biology***, 5(1), 678. https://doi.org/10.1038/s42003-022-03584-6

Fang, D., Cui, K., ***Cao, Y.***, Zheng, M., Kawabe, T., Hu, G., Khillan, J. S., Li, D., Zhong, C., Jankovic, D., Sher, A., Zhao, K., & Zhu, J. (2022). Differential regulation of transcription factor T-bet induction during NK cell development and T helper-1 cell differentiation. ***Immunity***, 55(4), 639–655.e7. https://doi.org/10.1016/j.immuni.2022.03.005

2021 Ku, W. L., Pan, L., ***Cao, Y.***, Gao, W., & Zhao, K. (2021). Profiling single-cell histone modifications using indexing chromatin immunocleavage sequencing. ***Genome Research***, 31(10), 1831–1842. https://doi.org/10.1101/gr.260893.120

Huang, F., ***Cao, Y.***, Wang, C., Lan, R., Wu, B., Xie, X., Hong, J., Fu, L., & Wu, G. (2021). PNMA5 promotes bone metastasis of non-small-cell lung cancer as a target of BMP2 signaling. ***Frontiers in Cell and Developmental Biology***, 9, 678931. https://doi.org/10.3389/fcell.2021.678931

2020 Qiu, X., Ma, F., Zhao, M., ***Cao, Y.***, Shipp, L., Liu, A., Dutta, A., Singh, A., Braikia, F. Z., De, S., Wood, W. H., III, Becker, K. G., Zhou, W., Ji, H., Zhao, K., Atchison, M. L., & Sen, R. (2020). Altered 3D chromatin structure permits inversional recombination at the IgH locus. ***Science Advances***, 6(33), eaaz8850. https://doi.org/10.1126/sciadv.aaz8850

Xia, X., Chen, X., Wu, G., Li, F., Wang, Y., Chen, Y., Chen, M., Wang, X., Chen, W., Xian, B., Chen, W., ***Cao, Y.***, Xu, C., Gong, W., Chen, G., Cai, D., Wei, W., Yan, Y., Liu, K., Qiao, N., Zhao, X., Jia, J., Wang, W., Kennedy, B. K., Zhang, K., Cannistraci, C. V., Zhou, Y., & Han, J.-D. J. (2020). Three-dimensional facial-image analysis to predict heterogeneity of the human ageing rate and the impact of lifestyle. ***Nature Metabolism***, 2(9), 946–957. https://doi.org/10.1038/s42255-020-00270-x

2019 Chen, X., Xu, C., Hong, S., Xia, X., ***Cao, Y.***, McDermott, J., Mu, Y., & Han, J.-D. J. (2019). Immune cell types and secreted factors contributing to inflammation-to-cancer transition and immune therapy response. ***Cell Reports***, 26(7), 1965–1977.e4. https://doi.org/10.1016/j.celrep.2019.01.080

2018 Xu, C., Ai, D., Shi, D., Suo, S., Chen, X., Yan, Y., ***Cao, Y.***, Zhang, R., Sun, N., Chen, W., McDermott, J., Zhang, S., Zeng, Y., & Han, J.-D. J. (2018). Accurate drug repositioning through non-tissue-specific core signatures from cancer transcriptomes. ***Cell Reports***, 25(2), 523–535.e5. https://doi.org/10.1016/j.celrep.2018.09.031

2015 Chen, W., Qian, W., Wu, G., Chen, W., Xian, B., Chen, X., ***Cao, Y.***, Green, C. D., Zhao, F., Tang, K., & Han, J.-D. J. (2015). Three-dimensional human facial morphologies as robust aging markers. ***Cell Research***, 25(5), 574–587. https://doi.org/10.1038/cr.2015.36

Huang, Y., Yu, X., Sun, N., Qiao, N., ***Cao, Y.***, Boyd-Kirkup, J. D., Shen, Q., & Han, J.-D. J. (2015). Single-cell-level spatial gene expression in the embryonic neural differentiation niche. ***Genome Research***, 25(4), 570–581. https://doi.org/10.1101/gr.181966.114

2014 Hong, S., Huang, Y., ***Cao, Y.***, Chen, X., & Han, J.-D. J. (2014). Approaches to uncovering cancer diagnostic and prognostic molecular signatures. ***Molecular & Cellular Oncology***, 1(2), e957981. https://doi.org/10.4161/23723548.2014.957981

SOFTWARE

2022-Curr. astroBoy

An automated, AI-assisted next-generation sequencing analysis platform for in-house and public datasets on a single-node server. Deployed at NHLBI/NIH;

processed over 530 billion reads from more than 10,000 samples across 900 analysis jobs since December 2022.

2023-Curr glitter
An in-house web application for analyzing and visualizing single-cell RNA-seq data to support collaborative research.

2023-Curr VINCE
A Python package for nucleosome analysis using single-cell MNase-seq data, currently under active development and optimization.

2026–Curr. cLoops3
Chromatin interaction analysis software under development since 2026, with assistance from Codex for code enhancements.

2026 Ryder
A Python package for epigenomic normalization using internal reference regions and variable-feature detection. Accepted and published in *Bioinformatics* (2026).

2021 cLoops2
A Python package for comprehensive analysis of chromatin interaction data, including Hi-TrAC, HiChIP, and Hi-C (*Nucleic Acids Research*, 2022).

2019 TOWN
A Keras-based Python package for automated counting of live *C. elegans* in high-throughput bright-field microscopy images using deep learning. Developed for in-house drug screening to identify compounds that improve worm survival.

2017 cLoops
A Python package for accurate and flexible loop calling from 3D genomic data (*Bioinformatics*, 2020).

PEER REVIEW

Nature Communications, Genome Biology, Bioinformatics, Communications Biology, BMC Bioinformatics, BMC Genomics, F1000Research, Frontiers in Immunology, Frontiers in Bioinformatics, Genes, and Cells.

EDITORIAL SERVICE

Frontiers in Immunology, Frontiers in Medicine, and Frontiers in Epigenetics and Epigenomics

AWARDS AND HONORS

2024 **NHLBI Director's Award for Innovation**
Discovery of a novel regulatory element for IFN-γ expression (*Immunity*, 2023).

MENTORING

2026- Nikita Ostro
NIH Postbac Program
Co-mentoring in epigenomic and Hi-TrAC chromatin interaction analysis.

2026- Anning Cui
NIH Postbac Program
Co-mentoring in epigenomic and single-cell nucleosome data analysis.

2025- Paul Elizalde
NIH-JHU Graduate Partnerships Program (GPP) Ph.D. student
Co-mentoring in bioinformatic analysis of epigenomic data.

2025- Tuan Vinh
NIH Oxford-Cambridge Scholars Program Ph.D. student
Co-mentoring in bioinformatics tool development and chromatin interactions.

2024,2023 Hannah Zhu
University of Maryland; NIH Summer Internship Program.
Mentored in STARR-seq and Hi-TrAC data analysis to study enhancer–enhancer interactions.

2024 Emily Yu
The Bryn Mawr School; High School Senior Project.
Mentored in single-cell RNA-seq analysis; currently at Yale University.

2019-2020 Jonathan Perrie
NIH Postbac Program. Co-mentored in bioinformatics.
Subsequently enrolled in the bioinformatics Ph.D. program at UCLA.

2015-18 Yingying Zeng
Partner Institute for Computational Biology; junior master's student. Co-mentored in bioinformatics and collaborated on research investigating the epigenetic features of MIR retrotransposons. Co-first-authored a publication and subsequently enrolled in the Ph.D. program at Nanyang Technological University, Singapore.

2016-18 Zhaoxiong Chen
Partner Institute for Computational Biology; junior Ph.D. student. Co-mentored in bioinformatics and collaborated on optimizing the cDBSCAN algorithm and cLoops. Co-first-authored a publication.

2016-18 Daosheng Ai
Partner Institute for Computational Biology; intern. Co-mentored in bioinformatics.
Co-first-authored a publication and subsequently enrolled in the Ph.D. program at Peking University, China.

TALKS

2026 Bioinformatics Resources for Epigenomic Data Analysis. Guest lecture invited by Prof. Gangqing Hu, West Virginia University School of Medicine (Zoom, USA).
2025 TriLab Bioinformatics Core, NIDDK, Bethesda, USA
2024 Bioinformatics and Scientific Programming Core, NICHD, Bethesda, USA
2024 Invited by Dr. Gordon L. Hager, NCI, Bethesda, USA
2023 Invited by Prof. Lei Hou, Boston University, Boston, USA
2023 DECODE Seminar Series, NIH, Bethesda, USA
2017 Epigenetics Retreat, Ningbo, China

CONFERENCES & POSTERS

2025 NIH/FDA Immunology Interest Group Retreat, Washington, DC, USA
2024 NCI Histone Modifications and Chromatin Structure Symposium, Bethesda, USA
2024 Epigenetics & Chromatin, Cold Spring Harbor Laboratory, New York, USA
2024 NHLBI Fellow Festival, NIH, Bethesda, USA
2024 NIH/FDA Immunology Interest Group Retreat, Washington, DC, USA
2023 Genome Architecture in Cell Fate and Disease, Ventura, California, USA
2023 NHLBI Fellow Festival, Bethesda, USA
2022 SCBA DC-Baltimore Chapter Annual Scientific Symposium, University of Maryland, USA
2019 4D Nucleome Annual Meeting, Washington, DC, USA
2018 Chromatin, Epigenetics & Transcription, Suzhou, China
2017 First Chromatin Biology Conference, Shenzhen, China
2016 3rd International Symposium on 3D Genomics, Huazhong Agricultural University, Wuhan, China
2015 Otto Warburg Summer School, Max Planck Institute for Molecular Genetics, Berlin, Germany

TECHNICAL SKILLS

- Analysis of epigenomic, 3D genome, and single-cell data (Hi-C, HiChIP, ATAC-seq, ChIP-seq, MNase-seq, Hi-TrAC, RNA-seq, Micro-C, 10x single-cell RNA-seq, 10x ATAC + gene expression multiomics, STARR-seq, RRBS, EM-seq, and WES), including custom computational methods and complex data mining
- Python programming and package development
- CPU/GPU server setup and maintenance
- Web application development
- Application and development of deep learning and machine learning algorithms

REFERENCES

Dr. Keji Zhao	Distinguished Investigator, NHLBI/NIH	zhaok@nhlbi.nih.gov	301-496-2098
Dr. Paul P. Liu	Senior Investigator, NHGRI/NIH	pliu@nhgri.nih.gov	301-402-2529
Dr. Jinfang Zhu	Senior Investigator, NIAID/NIH	jfzhu@niaid.nih.gov	301-402-6662